Journal of Materials Processing Technology 209 (2009) 4465–4470

Contents lists available at ScienceDirect

Journal of Materials Processing Technology

journal homepage: www.elsevier.com/locate/jmatprotec

Effect of roll gap adjustment on exit cross sectional shape in groove rolling—Experimental and FE analysis

S.M. Byon [a], D.H. Na [b], Y. Lee [b,*]

[a] Department of Mechanical Engineering, Dong-A University, Busan 604-714, Republic of Korea
[b] Department of Mechanical Engineering, Chung-Ang University, Seoul 156-756, Republic of Korea

ARTICLE INFO

Article history:
Accepted 6 October 2008

Keywords:
Roll gap adjustment
Finite element analysis
Roll wear
Oval groove roll
Round groove roll

ABSTRACT

We machined work roll with groove worn down as well as groove with no wear. We then performed a pilot hot rod rolling test at temperature of 1000 °C using plain carbon steel (0.1% C) as the roll gap decreases from reference roll gap (6.5 mm) to 3.5 mm. To understand better the effect of roll gap (i.e., section height) adjustment on the exit cross sectional area (ECSA) variation of workpiece in a two-stand groove rolling process with wear is considered, we carried out a series of three dimensional finite element analysis. Results reveal that variation of ECSA is almost linearly proportional to roll gap change while the roll gap decreases from reference roll gap (6.5 mm) to 3.5 mm. In oval groove rolling, the exit cross sectional shape and area predicted by FEA is in a good agreement with those measured. In round groove rolling, however, some deviations between FEA and experiment are observed because of roll groove geometry coupled with cross sectional shape of incoming workpiece. In the two-stand groove rolling, the effect of roll gap adjustment at each stand on the exit cross section of workpiece is somewhat different, in comparison with single-stand groove rolling. The roll gap adjustment at the previous stand has a more influence on the ECSA of workpiece than that of the next stand.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

In continuous hot rod rolling process, billets are processed into rod with acceptable dimensional tolerance as they pass through the stands (i.e., a set of grooved rolls installed at rolling machine). ECSA – exit cross sectional area of workpiece at each stand (or pass) – is progressively reduced with the cross sectional shape altered by the grooved roll.

Researchers in the field of rod rolling analysis have been interested in understanding how the ECSA at a stand varies during rolling. Note ECSA is inevitably changed with the increase of roll tonnage (i.e. the total number of tons that the produced rod products weigh) due to wear. This leads to change of mass balance between stands. If an amount of wear on work roll is small, the mass balance at the stands can then be maintained by the inter-tension controlling system installed in continuous rod or bar mill. If, however, the amount of wear on work roll is large enough, the roll gap (i.e., section height) of all stands should then be adjusted to keep the mass balance. Therefore, a study of roll wear and roll gap adjustment have been highly desirable to deal with this problem.

Many researchers have studied macroscopic behavior of roll wear in hot rolling process experimentally or numerically. As experimental investigation, Sachs et al. (1961) measured roll wear contour in flat rolling. They reported the roll force was the most dominating factors influencing roll wear and wear of work rolls progressed linearly with rolled tonnage. Oike et al. (1977) proposed a wear model of flat roll expressed as a function of roll force, strip width, roll contact length, reduction ratio, exit strip length, roll diameter and empirical coefficients. In terms of thermal hysteresis loop, some researchers (Williams and Boxall, 1965; Stevens et al., 1971) related the plastic strain amplitude with the roll wear. They showed that the optimal water spray zone leaded to the increase of the roll life. Ryu et al. (1992) defined the thermal fatigue damage parameter as a function of the plastic strain amplitude and the number of revolution and showed that the parameter can be effectively used as a measure of evaluating the severity of roll wear. Spuzic et al. (1994) surveyed accumulated knowledge on how the roll wear occurs in hot rolling process and provided a direction of experimental method of roll wear. Based on experimentally measured wear profile of roll groove (oval and round), Byon and Lee (2007) recently presented a semi-analytical model for predicting the wear contour in oval-to-round (or round-to-oval) groove rolling as the roll tonnage increases. The proposed model was applied to the roughing train in the POSCO No. 2 Wire and Rod Mill.

For numerical approach, Kim et al. (2003) calculated the wear contour of work roll for a pilot bar rolling mill with an oval and round roll groove using finite element method coupled with Archard wear model (Archard, 1953). The roll wear profile was cal-

* *Corresponding author*. Tel.: +82 2 824 5256; fax: +82 2 814 9476.
E-mail address: ysl@cau.ac.kr (Y. Lee).

0924-0136/$ – see front matter © 2008 Elsevier B.V. All rights reserved.
doi:10.1016/j.jmatprotec.2008.10.045

culated at each deformation step in consideration of relative sliding speed and normal roll pressure at contact area. But they neither performed any experiment regarding roll wear nor investigated the relation between the amount of roll wear and the roll gap.

These studies, however, are simply confined to measuring or predicting roll wear contour. No study has been reported to examine a relationship between the amount of roll wear and the subsequent ECSA changes including roll gap adjustment when a certain amount of wear already exists on roll groove. To begin with, we designed wear contour based on the FE analysis results of Kim et al. (2003). The roll groove worn down was machined by CNC (Computerized Numerical Control) lathe. It should be noted that having wear in pilot rolling mill is almost impossible since at least a few hundreds tons of material should be rolled in order for the wear profile to be detected clearly. Pilot single-stand rod rolling test using groove worn down as well as groove with no wear has then been performed while the roll gap is adjusted. Incoming specimen (plain carbon steel) to the pilot rolling tester is of a round section.

Thus, we first investigated the relationship between the amount of roll gap adjustment and the increased ECSA due to wear in single pilot stand rolling test. At the same time, we conducted 3-D Finite Element Analysis. We then compared the measured exit cross sectional area and shapes with the computed ones when roll wear is considered. Note that rod or bar products are manufactured through series of stands and therefore understanding the effect roll gap adjustment on ECSA at the very next stand is very important. However, carrying out continuous rolling test is quite limited because of the rolling facilities for experiment.

Therefore, after confirming the usefulness of the 3-D FEA with the roll wear included, we examined the effect roll gap adjustment on ECSA at the very next stand in two-stand continuous rolling with the groove worn down using 3-D FEA.

2. Definition of the problem

2.1. Rod rolling process

Fig. 1 shows the schematic of deformation of workpiece during round-to-oval-to-round groove rolling. It means that workpiece with a round cross section is rolled in oval (shaped) roll groove and subsequently the one with an oval cross section is rolled in round (shaped) roll groove. Then, round shaped cross section of workpiece finally produced.

2.2. Roll wear and roll gap adjustment

Fig. 2 shows the geometrical designation of initial roll groove and wear contour for oval and round groove. The wear contour is based on the results of finite element analysis for oval and round groove rolling (Kim et al., 2003). Upper roll groove is depicted for expediency. All units are in millimeter. For the oval groove, the radius of curvature of initial groove is R_1 and half pass height is $H_p/2$. R_0 represents the radius of curvature oval roll groove worn down. Maximum wear occurs at the center of roll groove.

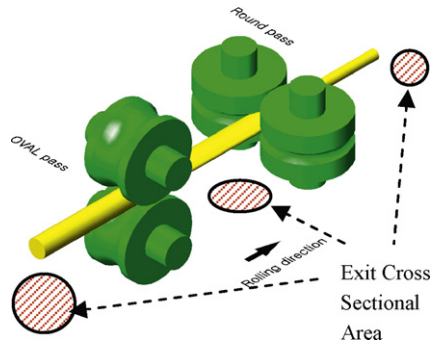


Fig. 1. Schematic of deformation of workpiece in round-to-oval-to-round groove rolling.

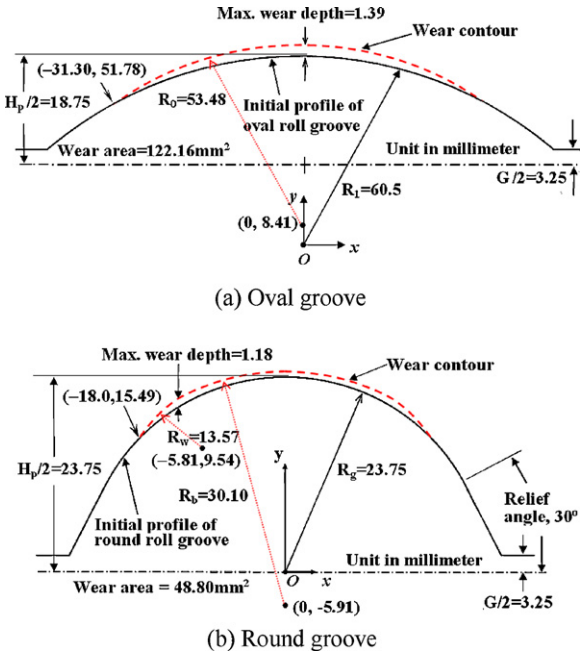


Fig. 2. Geometrical designations for initial roll groove and wear contour. (a) oval groove and (b) round groove.

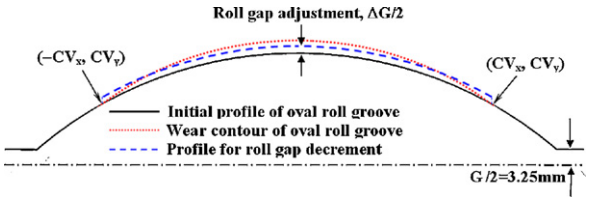


Fig. 3. Concept for roll gap adjustment, ΔG. (CV_x and CV_y denote the separating point where the oval roll groove and workpiece are separated each other.)

In case of round groove, wear contour have two radius of curvature, R_b and R_w. The starting point of R_b is on y-axis but that of R_w is neither on the x-axis nor y-axis. Maximum wear occurs at the region what is called 'roll shoulder' which is located between center of roll groove and separating point that the roll groove stops contacting the workpiece at the exit position. This is because the effect of grinding which occurs between roll and workpiece on wear is most significant at the roll shoulder region (Kim et al., 2003).

To reduce the increased ECSA owing to groove wear up to a base ECSA of workpiece, roll gap should be adjusted. Fig. 3 shows the concept of roll gap adjustment, ΔG in oval groove. Configuration of lower roll groove is omitted for brevity. Roll gap adjustment implies the amount of section height that has to be reduced when wear occurs. That is, the inside area surrounded by initial profile (marked in solid line) and wear contour (marked in dotted line) is compensated by the area marked in dashed line.

3. Experiment

3.1. Pilot rolling mill and wear contour

Fig. 4 shows a roll body with a round groove and two different oval grooves, and initial specimens with diameter of 60 mm used in the pilot rolling test. DCI (Ductile Casting Iron) roll of 310 mm

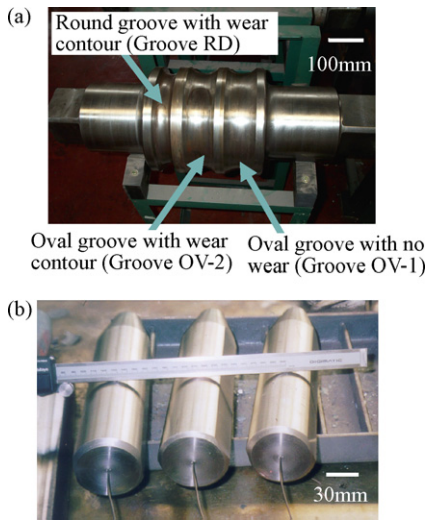


Fig. 4. (a) Roll with one round groove and two oval grooves and (b) initial specimens with diameter of 60 mm used in the rolling test.

diameter was used. Initial profile and wear contour of roll groove were machined by CNC lathe based on groove data in Fig. 2. Rolling speed is set at 34 rpm. Plain low carbon steel (0.1% C) specimen was used. The initial specimen was heated up to 1030 °C in an atmosphere of nitrogen and rolled at 1000 °C without lubrication. To measure the rolling temperature of the workpiece (specimen), K-type thermocouple was embedded in 40 mm deep holes drilled in the tail ends of the specimen.

3.2. Specimen

Eight initial specimens in total were used. The four initial specimens were rolled directly in the oval roll groove worn down (Groove OV-2 in Fig. 4(a)) as the roll gap was reduced from 6.5 to 5.5, 4.5 and 3.5 mm. To make initial specimen for the round groove with wear (Groove RD in Fig. 4(a)), we rolled the four initial specimens left in the oval groove with no wear (Groove OV-1 in Fig. 4(a)). This rolling leads us to have four workpieces with oval cross sectional shape and subsequently these workpieces were used as initial specimens for round groove worn down. The workpiece with the oval cross sectional shape was rolled in the round groove after it was rotated 90° about its length direction. Entry guides were installed in front of the round roll groove to minimize sideway bending of specimen.

3.3. Measurement of exit cross sectional area (ECSA)

The specimens after oval groove rolling and round groove rolling were cooled in air, respectively. The cross sections with 30 mm thickness were obtained by cutting the middle part of workpiece (rolled specimen) to the length (rolling) direction. Then, milling machine smoothed the cross section of workpiece. Finally, the coordinates of surface profile were obtained by using the surface profile reading program followed by scanning the cross section of workpiece.

4. Finite element analysis

We performed a series of finite element analysis (FEA) to understand better the change of ECSA of workpiece when wear exists on the roll groove. For this purpose, single-stand rolling analysis which coincides with the pilot rolling test condition is carried out. Then, two-stand rolling, corresponding to the continuous rod rolling, is modeled and analyzed. For FE simulation, a commercial FE code, ABAQUS® which has a good capability in the analysis of the nonlinear behavior of severe elastic–plastic deformation is employed.

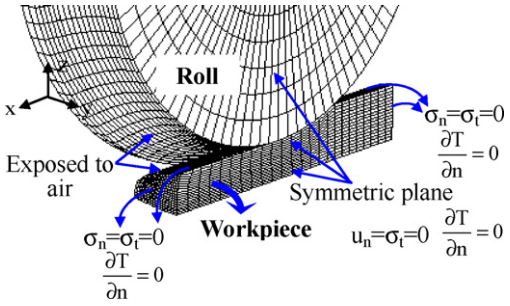


Fig. 5. FE meshes and boundary conditions used in the single-stand rolling analysis. (Symbolic parameters are described in Table 1.)

Table 1
Description of parameters on FE analysis.

Symbols	Meanings
u_n	Normal displacement
u_t	Tangential displacement
σ_n	Normal surface traction
σ_t	Tangential surface traction
T	Temperature
$\partial T/\partial n$	Temperature gradient

4.1. Single-stand rolling analysis

Fig. 5 shows the mesh and boundary conditions to analyze the single-stand rolling. The parameters on FE analysis are listed in Table 1. We adopted a quarter FE model to reduce the computational time. To evaluate the interdependency among the mechanical behavior of the workpiece, thermal behavior of the workpiece, and thermal behavior of the roll, the coupled FE analysis was performed. The workpiece section and the roll section were taken as the analysis domain. Mechanical boundary conditions were described for the workpiece and thermal boundary conditions for the roll as well as the workpiece accordingly. We assumed that initial temperature of incoming workpiece was 1000 °C. Since working temperature is very high, the strain hardening effect of initial workpiece was neglected. It was also assumed that tensions were not applied to the front end and rear end of workpiece.

4.2. Continuous two-stand rolling analysis

Geometry of work roll, workpiece and their meshes in continuous two-stand rod rolling process are shown in Fig. 6. To save the computational time, a quarter analysis domain and isothermal deformation assumption were taken. The analysis model is based on the H (Horizontal)–V (Vertical) type continuous rolling. Thus,

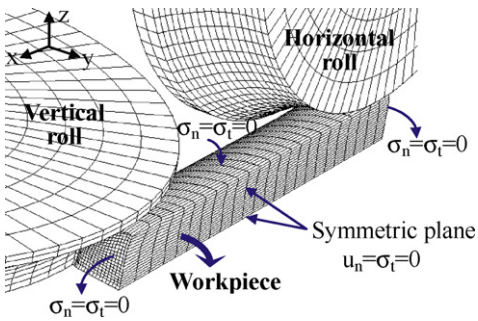


Fig. 6. FE meshes and boundary conditions used in the continuous two-stand rolling. (Symbolic parameters are described in Table 1.)

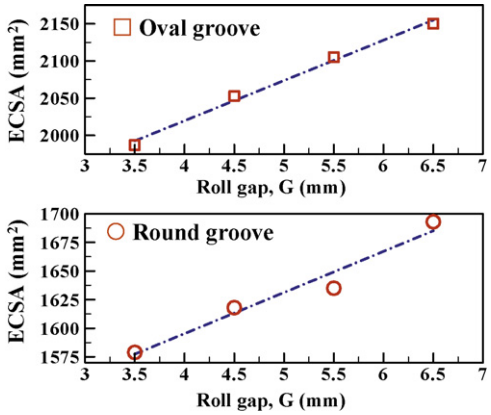


Fig. 7. Measured ECSA as a function of roll gap decrement in oval and round groove.

the workpiece is forced to deform by upper and lower rolls at the first stand and right and left rolls at the second stand. To achieve the mass balance of workpiece between two stands, the roll velocity of the second stand was regulated based on the mass flow rate of the first stand. In this calculation, the concept of mean roll radius (Lee, 2001) was adopted. The tensions at the front and rear end of workpiece were not exerted.

5. Results and discussion

5.1. Exit cross sectional areas (ECSA) measured

Fig. 7 shows the variation of ECSA measured at the oval groove and round groove, respectively, while roll gap varies from 6.5 to 3.5 mm. The variation of ECSA is nearly linearly proportional to roll gap adjustment. In the case of oval groove, original ECSA at the reference state (i.e. roll groove with no wear and reference roll gap: 6.5 mm) is 2060 mm^2. In the reference roll gap, however, if the initial workpiece with round cross section was rolled in the oval groove worn down, the ECSA increased to 2150 mm^2. Hence we should decrease the roll gap, i.e., section height, to make the ECSA 2060 mm^2 approximately. We can observe the ECSA decreases as the roll gap decreases. At this point, we may guess if the roll gap is slightly larger than 4.5 mm, the ECSA obtained from oval groove worn down might reach the reference ECSA, 2060 mm^2.

Fig. 7 also shows measured ECSA in the round groove as workpiece with oval cross section is rolled in the round groove worn down and round groove with no wear. A similar tendency is observed in comparison with the oval groove rolling. At the reference roll gap, i.e., 6.5 mm, ECSA in the round groove with no wear, the ECSA is 1628 mm^2. We can then find that roll gap which can produce the reference ECSA might exist between 4.5 and 5.5 mm.

5.2. Comparison of ECSA—experiment and computation

Table 2 summarizes the ECSA calculated from FE analysis when wear is already present on the oval groove and round groove. Error indicates the difference between measured ECSA and computed one.

Table 2
Predicted ECSA at oval groove and round groove.

Roll gap (mm)	Oval groove worn down		Round groove worn down	
	ECSA (mm^2)	Error (%)	ECSA (mm^2)	Error (%)
6.5	2153	0.1	1704	0.7
5.5	2109	0.2	1658	1.4
4.5	2071	0.9	1632	0.8
3.5	2001	0.7	1614	2.2

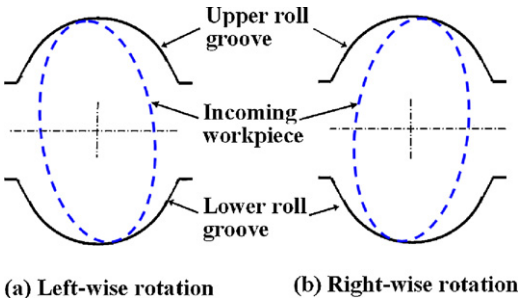


Fig. 8. Possible rotation of incoming workpiece and round roll groove at the very biting position.

The difference between measured ECSA and computed one is expressed as error. In the case of oval groove, the error is less than 1.0%. On the other hand, we can observe the error significantly increases in round groove. Especially, the error at roll gap of 3.5 mm reaches 2.2%. This is because the incoming workpiece is somehow rotated along rolling direction during rolling. This is due to an unstable rolling attributed to a loose entry guider at the entrance of round roll groove.

Fig. 8 illustrates the cause of the unstable rolling at the round groove rolling. If the cross section of incoming workpiece is of a round shape, rolling with the oval groove is very stable. On the other hand, rolling in the round groove is unstable if incoming workpiece is of an oval cross section as shown in Fig. 8. This kind of problem occurs straightforwardly in round groove rolling.

Fig. 9 shows the exit cross sectional shapes of workpiece obtained by experiment are compared with those predicted by FE analysis while the roll gap of oval groove varies from 6.5 to 3.5 mm. Solid lines represent the computed exit cross sectional shape and dashed lines the experimentally measured one.

As shown in the figure, a good accord between experiments and calculations is noted. This result indicates that, in oval groove rolling, the shape and area of exit cross section obtained from FE simulation is in a good agreement with those from experiment. It is attributable to the stable rolling condition of oval groove rolling as explained in Fig. 8 and subsequently experimental conditions are matched well with FE simulation condition.

Fig. 10 illustrates ECSA of workpiece obtained by experiment are compared with those calculated by FE simulation while the roll gap of round groove varies from 6.5 to 3.5 mm. In comparison with oval groove, the exit cross sectional shapes measured do not coincide in a good way with those calculated by FE analysis. This is because, in the groove rolling test, workpiece with oval cross section inside the round groove is rotated. In the meantime FE analysis did not consider this during simulation. This is observed in the measured exit cross sectional shapes in Fig. 10(a) and (d), especially.

5.3. Effect of roll gap change on ECSA in two-stand groove rolling with roll groove worn down

To examine the effect of roll gap change on ECSA in a two-stand groove rolling with roll groove worn down, we took into account three cases as summarized in Table 3. Isothermal condition dur-

Table 3
Adjusted roll gap for FE analysis and variations of ECSA in the continuous two-stand rolling.

Cases	Adjusted roll gap (mm)		ECSA (mm^2)	
	1st stand (oval groove)	2nd stand (round groove)	1st stand (oval groove)	2nd stand (round groove)
Case-A	4.64	5.28	2061	1676
Case-B	6.50	5.28	2168	1726
Case-C	4.64	6.50	2063	1712

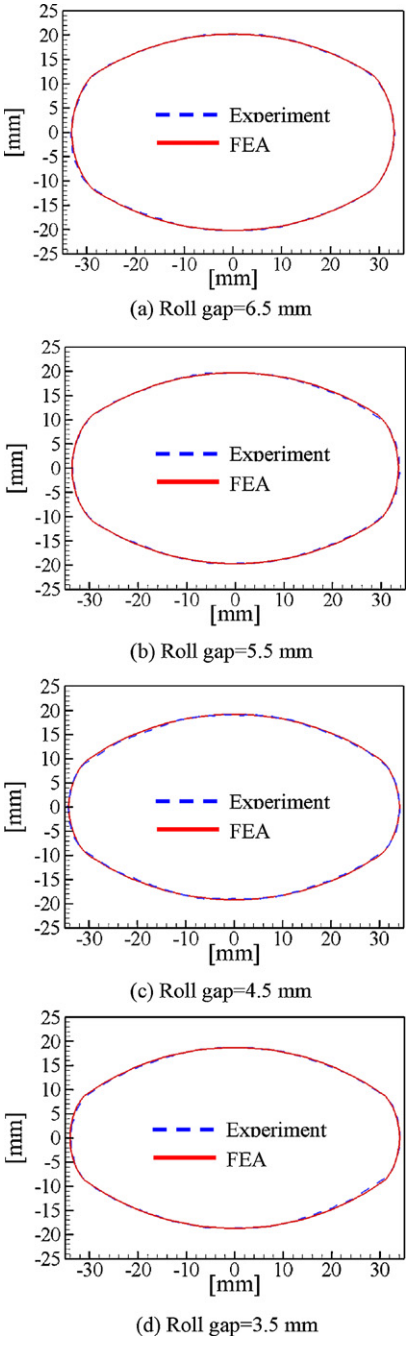


Fig. 9. Predicted and measured exit cross sectional shape in oval groove rolling.

ing rolling is assumed. Note that heat flow at the interface of roll and workpiece, convection at the workpiece surface and radiation has been considered for single-stand rolling analysis but not for the two-stand rolling analysis since ABAQUS® can not perform the elastic–plastic deformation analysis coupled heat transfer for such a multi-stand rolling process. To check the validity of the assumption, we carried out FE simulation of elastic–plastic deformation inside roll groove under isothermal and non-isothermal condition for the case of the single-stand groove rolling. A very small difference, say, 0.2–0.3%, has been found. Hence we can perform two-stand rolling analysis under isothermal condition.

Case-A indicates the roll gap adjusted to yield the reference ECSA (2060 mm^2 for round groove and 1628 mm^2 for oval groove) as much as it can. To observe the effect of roll gap adjustment in terms of ECSA at the previous stand on the very next stand, two

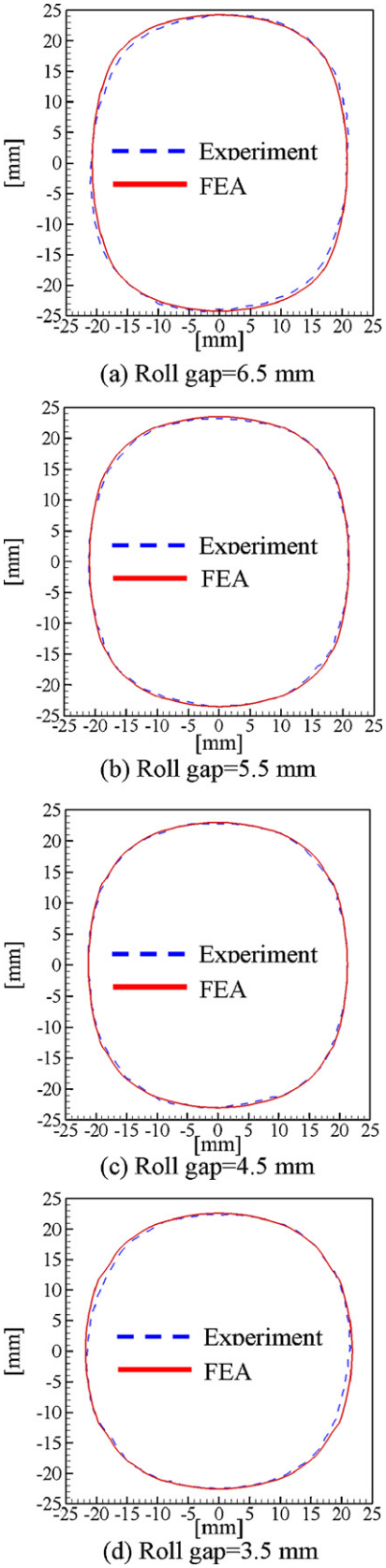


Fig. 10. Predicted and measured exit cross sectional shape in round groove rolling.

cases (Case-B and Case-C) were considered. Case-B stands for that the roll gap of 1st stand is unchanged (i.e., remains at reference roll gap) but that of 2nd stand is adjusted. Opposite roll gap adjustments is applied to Case-C.

We can observe when we use roll gap adjustments of the 1st and 2nd stand obtained from single-stand rolling analysis, ECSA at round groove in the two-stand rolling is slightly higher than for the single-stand rolling. This is attributable to the amount of wear included in the cross sectional area at the exit of oval groove.

It is interesting to note that we have more reduced ECSA at round groove when roll gap at the 1st stand is adjusted rather than at the 2nd stand. This is because the ECSA at oval groove is much larger than that at the round groove. Hence adjusting roll gap at the previous stand is recommended if one want to reduce ECSA at the very next stand.

6. Concluding remarks

We have performed a pilot hot groove rolling test which has roll groove worn down as well as roll groove with no wear to investigate the relation between variations of roll gap (i.e., section height) and ECSA in oval-to-round and round-to-oval groove rolling. We also carried out FE analysis to examine the effect of roll gap adjustment on the exit cross sectional shape and area in two-stand rolling as well as single-stand rolling when a certain amount of wear on roll groove already exists. The conclusions are summarized as follows:

The ECSA measured is, to some extent, linearly dependent on the roll gap adjustment for the given round and oval groove wear profiles. The exit cross sectional shape and area calculated by FE simulation is in a good agreement with those measured for oval groove rolling, but for round groove rolling some deviations (∼2.2%) are noted because of rotation of workpiece during rolling. The ECSA increased due to wear at the previous stand does not influence much the ECSA at the very next stand.

Acknowledgement

This study was supported by research funds from Dong-A University.

References

Archard, J.F., 1953. Contacts and rubbing of flat surfaces. J. Appl. Phys. 24, 981–988.
Byon, S.M., Lee, Y., 2007. Experimental and semi analytical study of wear contour of roll groove and its applications to rod mill. ISIJ Int. 47, 1006–1015.
Kim, D.H., Lee, Y., Yoo, S.J., Choo, W.Y., Kim, B.M., 2003. Prediction of wear profile of roll groove in rod rolling using an incremental form of wear model. Proc. Inst. Mech. Eng. Part B: J. Eng. Manuf. 217, 111–126.
Lee, Y., 2001. An analytical study of mean roll radius in rod rolling. ISIJ Int. 41, 1414–1416.
Oike, Y., Okubo, I., Hirano, H., Umeda, K., 1977. Tetsu-to-Hagane 63, S222 (in Japanese).
Ryu, J.H., Kwon, O., Lee, P.J., Kim, Y.M., 1992. Evaluation of the finishing roll surface deterioration at hot strip mill. ISIJ Int. 32, 1221–1223.
Sachs, G., Latorre, J.V., Chakko, M.K., 1961. Roll wear in finishing trains of hot strip mills. Iron Steel Eng. 38, 71–91.
Spuzic, S., Strafford, K.N., Subramanian, C., Savage, G., 1994. Wear of hot rolling mill rolls: an overview. Wear 176, 261–271.
Stevens, P.G., Ivens, K.P., Harper, P., 1971. Increasing work roll life by improved roll cooling practice. J. Iron Steel Inst. 209, 1–11.
Williams, R.V., Boxall, G.M., 1965. Roll surface deterioration in hot strip mills. J. Iron Steel Inst. 203, 369–377.